Midterm Progress check

Member	Email	Presentation Contribution	Implementation Contribution
Priyanshu Srivastava	ps5e25@soton.ac.uk	Societal Impact	tokenisation + lexers for each language
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Xinxie Li	xl9n25@soton.ac.uk	Demo	Interface for the Demo

Topic: Program Text Analysis

Artefact

We're building a command-line lexical analyzer in Python that processes code written in our own simplified programming language (based on Python/Lua syntax). The tool accepts source code using English, Spanish, or Mandarin keywords as input. Internally, it tokenizes the code into a language-neutral representation, then outputs either the tokenized form (showing token types and values) or translates the code into one of the other supported languages. For example, you could write a program using Spanish keywords, pass it through the lexer with a translation flag, and get back the equivalent code in English or Mandarin. This demonstrates how lexical analysis can be language-agnostic while maintaining the same underlying token structure across different natural languages.

Motivation

Most programming languages are almost entirely English-centric. Our goal is to show that the basic principles of tokenisation and lexical parsing can be extended to handle multiple natural languages. By building a lexer that processes English, Spanish, and Mandarin keywords interchangeably, we demonstrate one way that the language barrier in programming can be alleviated to make programming more accessible for more people.

Societal Impact

English language dominance in programming creates barriers for billions of potential developers. A Spanish-speaking student in Latin America or a Mandarin speaker in rural China has to learn English and programming at the same time, which doubles the cognitive load. While you only really need a subset of English vocabulary to program, this still creates a significant barrier to entry for many people around the world.

Non-English programming languages do exist (the Chinese language 文言 (Wenyan) transpiles to JavaScript for example), but generally lack widespread adoption, interoperability and shareability between linguistic groups. Our multilingual lexical analyzer demonstrates a different approach: supporting Spanish and Mandarin keywords alongside English within a single language, while enabling translation between them. This maintains shareability and interoperability of code across linguistic groups while lowering barriers to entry for technical education in non-English regions and widening the global tech talent pool.

Team Contributions

Our system centers on a language-neutral token representation. Priyanshu is building a config file format that maps regex patterns to token types for each language (e.g., "si" → IF_TOKEN, "if" → IF_TOKEN) plus the parser that integrates this into the codebase. Toby is developing the core lexer that uses these configs to tokenize input, the translation logic that converts tokens back into any supported language, and the overall project architecture. Ruicheng is creating the testing framework that validates tokenization consistency across languages and ensures translation maintains semantic equivalence. Xinxie is building the CLI that handles user interaction and orchestrates the pipeline. The flow works like this: user input goes through the CLI to the lexer, which uses the config system to generate tokens that either get displayed directly or translated into another language, with tests validating consistency throughout.